

SUPPLEMENT TO “CERTIFIED BELLMAN OPERATORS FOR COSTLY POLICY REVISION”

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This supplement gives exact primitive contracts, complete computational accounting, the transfer budget, a discounted infinite-horizon tail result, and a theorem-to-checker map. The prior article and technical supplement are retained without modification in a separate historical volume. Numerical proposal, independent arithmetic verification, structural mathematical assumptions, and publication provenance are distinguished throughout.

1. CONTROLLED-MODEL PRIMITIVE CONTRACT AND COMPLETE ACCOUNTING

The controlled cases use the exact generator in `regret.py`, independently reconstructed by `audit.py`. Regimes are $i = 0, \dots, n - 1$. For each regime and action, four integer weights are drawn, in that order, from

$$(\max\{1, 38 - 10a\} + u_1, 36 + u_2, 12 + 7a + u_3, 2 + 11a + u_4), \quad u_j \in \{0, 1, 2, 3\}.$$

Normalize these weights and assign them to successors $\max(0, i - 1)$, i , $\min(n - 1, i + 1)$, and $n - 1$, combining coincident successors. The seeded standard-library pseudorandom generator is advanced in regime/action order. The stage reward is

$$r_{ia} = \frac{i}{n - 1} - c_a - \frac{av}{100000}, \quad c = (0, 9/100, 8/25, 11/20), \quad v \in \{1, \dots, 19\},$$

where v is drawn immediately after the four transition weights. The installed action is zero for $i \geq \lfloor n/2 \rfloor$ and is the seed modulo two otherwise. The implementation cost is $(1 + i/(n - 1))$ times the indicator of a realized departure. The terminal operating value is $i/\lfloor 2(n - 1) \rfloor$, and the initial law is uniform. Transitions, rewards, and costs are constant over dates; Bellman disadvantages generally are not. All primitives and every declared seed are retained in the protocol and compressed proof objects.

The original registration is commit 18cbcf41a4c35660be824f6984f68a1d38288a9. The disclosed arithmetic amendment is 950b1f1ac2ebaf974d4ef528add8b2c2e4ce72b. The amendment retains all cases and changes only certificate accumulation and its theoretical diagnostics. Its numerical constructor, independent reader, and runner hashes are checked against the final sources. This sequence is not called a fully code-frozen holdout. The old source and partial execution record remain in the R43 history directory.

TABLE I: Controlled audit: complete proposal and verification work

Case	Method	Coordinates	Variables	Nonzeros	Build	LP	Arithmetic	Check	Bytes
00	U	24	144	1102	0.005	0.006	0.011	0.015	1380
00	S	24	144	1150	0.005	0.004	0.010	0.017	1290
01	U	128	1040	8611	0.039	0.050	0.081	0.123	6804
01	S	128	1040	8867	0.040	0.072	0.084	0.130	7484
02	U	384	3408	29018	0.176	0.340	0.290	0.456	19833

Case	Method	Coordinates	Variables	Nonzeros	Build	LP	Arithmetic	Check	Bytes
02	S	384	3408	29786	0.159	0.367	0.297	0.481	19963
03	U	1152	10200	89840	0.570	9.077	0.940	1.574	56498
03	S	1152	10200	91376	0.590	27.514	0.969	1.580	56699
04	U	1024	9488	81406	0.612	2.182	0.915	1.573	52718
04	S	1024	9488	83454	0.634	2.032	0.938	1.599	52390
05	U	768	6488	57088	0.307	2.328	0.576	0.895	37114
05	S	768	6488	58112	0.315	1.935	0.632	0.955	40608
06	U	1536	13720	121698	0.852	30.052	1.000	1.951	2491
06	S	1536	13720	123746	0.870	30.052	1.002	1.955	2495
07	U	2048	19472	169793	1.635	30.078	1.608	3.468	3575
07	S	2048	19472	173889	1.641	30.077	1.673	3.541	3583

Times are seconds. Coordinates are $nT(m-1)$, not just nT . Each row uses one root relaxation, so its entire certified bound improvement is a one-node result; this is not a node-count scaling experiment. The LP has a 30-second cap. Model construction, exact constructor arithmetic, and independent verification are additional. Process high-water memory is recorded in the raw rows. Since the runner is sequential, that memory includes earlier allocations and cannot identify a causal per-method memory advantage. Comparing isolated peak memory would require separate fresh processes. No such comparison is asserted.

2. DIRECTED CERTIFICATE ARITHMETIC AND INDEPENDENT RECONSTRUCTION

Write the root relaxation as $\min c'z$ subject to $Ez = e$, $Az \leq b$, and $\ell \leq z \leq u$. Any rational vector y and any rational $\mu \geq 0$ give the lower bound

$$y'e - \mu'b + \sum_j \min\{h_j\ell_j, h_ju_j\}, \quad h = c - E'y + A'\mu.$$

Add the exact reference cost $\nu'H_0$ and truncate at zero. A floating-point LP success flag is neither necessary nor sufficient for this inequality. Numerical equality marginals and sign-corrected inequality marginals are merely proposals for y, μ .

Accumulating all distinct rational denominators before rounding is unnecessarily expensive. The implemented reader and constructor instead use denominator 2^{40} . They round each constant contribution downward. For each residual coefficient, they accumulate dyadic lower and upper enclosures $h_j^- \leq h_j \leq h_j^+$. The safe contribution is the downward-rounded minimum of

$$h_j^-\ell_j, h_j^-u_j, h_j^+\ell_j, h_j^+u_j.$$

Taking all four products is necessary when a savings coordinate can be negative. This procedure is valid for arbitrary proposed multipliers with the required sign, and keeps denominator growth under control. The complete lower endpoint, including reference cost, is retained as a rational number.

For the upper endpoint, the constructor repairs backward toward the reference action. It reserves a local inward margin before rounding nonreference probabilities down to denominator 2^{40} ; the unused probability is assigned to the reference action. At a fixed repaired continuation, the rounding-induced regret change is at most $(m-1)2^{-40} \max_a |q_a - q_{a^*}|$. If the reserved target leaves insufficient reference slack, the constructor uses the feasible reference action.

TABLE II
SUFFICIENT ROOT-WIDTH CONSTANTS FOR EVERY CONTROLLED MODEL

Case	ε	$\underline{d}$	$\lambda_+ - \lambda_-$	$\overline{E}_D$	$E_{S,0}$	Width bound
00	1/100	0.00371	603	0.0027	1.63	16.9
01	1/100	0.00061	1.2e+03	0.0167	19.8	75.1
02	1/100	0.000116	1.24e+04	0.126	1.53e+03	1.64e+03
03	1/100	0.000331	5.05e+03	0.0563	285	799
04	1/1000	2.2e-05	8.49e+04	0.00709	602	767
05	1/1000	0.000156	1.03e+04	0.000715	4.99	376
06	1/20	6.2e-05	2.99e+04	4.73	1.27e+05	1.28e+05
07	1/100	7.44e-05	2.53e+04	0.129	3.18e+03	4.04e+03

Values are descriptive decimal approximations of retained rational quantities. Error recursions are rounded upward to dyadic numbers. The sufficient bound concerns an exact minimizing LP policy and exact repair; a numerical proposal gap and endpoint arithmetic losses are not included in this theoretical column. Actual certified widths in the main table include those losses and can be larger when a solver stops. Constants are not suppressed when they are uninformative.

These are proposal choices. The independent reader ignores the repair routine and checks the serialized probabilities by the original operating and implementation Bellman equations. Every probability is nonnegative and sums to one, and every date/regime satisfies the original, unmodified allowance.

The independent endpoint reader imports neither the constructor nor NumPy, SciPy, or any optimizer. It reconstructs the operating optimum, reference policy, disadvantage and savings coefficients, all variable bounds, all product envelopes, and every cone and Bellman constraint. A further independent audit reconstructs the declared random primitive sequence and checks each complete proof against its registered parameters. It also independently recomputes the continuum weighted sums. These checks do not mechanically prove the analytic repair, rate, or disintegration theorems. Four retained mutations test rejection of a forged endpoint, negative action probability, negative inequality multiplier, and invalid transition mass.

3. ALL CONTROLLED CONTINUUM ENDPOINTS

The model has a persistent observed size $z \in [0, 1]$. Regime transitions are controlled but independent of size. Operating primitives are multiplied by $1 + z$ and implementation costs by $1 + z/2$. Thus the finite allowance is $0.01/(1 + z)$; it is not 0.01 after rescaling. The following table includes all 17 endpoints used by every displayed partition.

TABLE III: Every finite endpoint used in the atomic continuum certificate

Endpoint	Allowance	Lower	Upper	Width	Check (s)
0	0.01000000	7.765997	7.883084	0.117085733	0.054
1	0.00941176	7.822697	7.929826	0.107128306	0.054
2	0.00888888	7.872910	7.971275	0.098364582	0.053
3	0.00842105	7.918003	8.007721	0.089717171	0.053
4	0.00800000	7.958346	8.040570	0.082223429	0.054
5	0.00761904	7.994660	8.070364	0.075703281	0.053
6	0.00727272	8.027518	8.097495	0.069976171	0.054

Endpoint	Allowance	Lower	Upper	Width	Check (s)
7	0.00695652	8.057475	8.122294	0.064817828	0.054
8	0.00666666	8.084816	8.145021	0.060203701	0.054
9	0.00640000	8.109872	8.165848	0.055975509	0.054
10	0.00615384	8.132909	8.185072	0.052162255	0.054
11	0.00592592	8.154153	8.202063	0.047908675	0.054
12	0.00571428	8.173805	8.218688	0.044882388	0.054
13	0.00551724	8.192044	8.234190	0.042144623	0.054
14	0.00533333	8.209073	8.248540	0.039466645	0.054
15	0.00516129	8.224969	8.260548	0.035578105	0.054
16	0.00500000	8.239809	8.273341	0.033531010	0.054

Endpoint j corresponds to $z = j/16$. Allowances are printed downward for display only; the actual solve and checker use the exact rational allowance. Lower endpoints are rounded down and upper endpoints and widths up. The 2-, 4-, 8-, and 16-cell calculations reuse subsets of these predeclared endpoints. No new model or initial law is substituted when the partition changes.

For the general observable-fiber theorem, the bijectivity of the covariate evolution has an economic content: the label that selects a continuation is recoverable from the current observed state. Noninvertible maps can merge different labels. Selecting a different continuation for each merged label would add memory and would not prove the stated common-Markov result. Likewise, an action-dependent continuous-condition transition is not covered merely because it has finitely many successors. The original affine and nonlinear condition models are therefore retained as separate, open-width evidence.

4. EXACT SERVICE-ECONOMY CONTRACTS

Regimes are indexed by $i = 0, \dots, n - 1$. Demand, rewards, installed actions, costs, and initial masses are defined in the main article. Ties in reward maximization use the lowest action index. All inputs are rational. The three regime laws are as follows. In the banded law, $P_{i,i+1} = P_{i+1,i} = (1 + (i \bmod 3))/16$ and P_{ii} is the remaining row mass; other entries are zero. The cyclic law has $P_{ii} = 3/4$ and $P_{i,(i+1) \bmod n} = 1/4$. The dense law has $P_{ij} = 1/(2n) + (1/2)\mathbf{1}\{i = j\}$. Each is stochastic and independent of the action. These definitions, including the installed rule and initial distribution, are reconstructed independently by the checker.

The continuous state space consists of n disjoint copies of $[0, 1]^2$ equipped with their Borel sigma fields; the half-open convention assigns each modulo boundary a unique image. If $S_1(u, v) = (u + v^2 \bmod 1, v)$, then for every bounded Borel h , Fubini's theorem and translation invariance on the circle give $\int h(S_1(u, v)) du dv = \int h(u, v) du dv$. The analogous identity holds for $S_2(u, v) = (u, v + u^2 \bmod 1)$. Their composition preserves uniform measure. Conditional on each possible regime successor, this is precisely the pushforward identity required by the main aggregation theorem. Its proof concerns arbitrary bounded Borel test functions, not only partition indicators. Pointwise regime transition probabilities give the other aggregation identity. The transition from an individual state is nonetheless a finite sum of Dirac masses.

The within-cell density alternative is $1 + \zeta f(u)f(v)$, where $f(u) = 2u_1 - 1$ on $[0, 1)^2$. Since $\int f = 0$ and $\int |f| = 1/2$, it is a probability density for $|\zeta| \leq 1$, and its source average is the uniform destination law. For a fixed source, its L_1 difference from the reset density is $|\zeta f(u)|/2$. The supremum over sources is $|\zeta|/2$ (a supremum need not be attained on the half-open cell). This is the operator norm over test functions bounded by one. Probability total variation is

half this quantity. In both models the specified initial mixture uses exactly the same uniform reference measures; no approximate initial-law substitution is made.

5. COMPLETE WORK ACCOUNTING AND FINITE CERTIFICATES

TABLE IV: All structured cases: work and proof-object size

Case	nT	Variables	Nonzeros	LP (s)	Full LP (s)	Exact (s)	Check (s)	Bytes
1	64	256	538	0.0032	0.0036	0.0057	0.0065	2233
2	256	1024	2226	0.0063	0.0140	0.0266	0.0355	5648
3	1024	4096	9058	0.0191	0.1127	0.1546	0.2239	17114
4	4096	16384	36546	0.2618	1.2414	0.9772	1.6081	56086
5	64	256	496	0.0032	0.0036	0.0051	0.0063	1976
6	256	1024	2016	0.0079	0.0119	0.0248	0.0333	5123
7	1024	4096	8128	0.0326	0.0753	0.1440	0.2201	15921
8	256	1024	5376	0.0074	0.0227	0.0391	0.0394	5135
9	64	256	538	0.0032	0.0038	0.0057	0.0064	2188
10	256	1024	2226	0.0064	0.0140	0.0267	0.0348	5616
11	1024	4096	8128	0.0325	0.0781	0.1515	0.2075	15251
12	256	1536	3250	0.0079	0.0154	0.0321	0.0416	6577

“LP” is the eliminated regret program; “Full LP” retains operating and implementation Bellman values with the same common policy. Both use HiGHS with presolve and the same numerical feasibility tolerances and 60-second solver limit. They solve the same rational mathematical problem, but their floating-point objective estimates are not the certificates. The exact endpoints use a separately evaluated rational policy and rational equality-multiplier residual correction. A 10^{-9} inward allowance is a proposal device only. Policy and dual multipliers are rounded on denominator 10^{12} ; the checker always uses the original declared allowance. Every row’s exact repair fraction was zero in the recorded execution; the feasibility assertion follows from exact evaluation, not from that observation or the floating-point solver status.

The raw result file additionally reports model preparation time, solver iterations, original formulation dimensions, and process high-water resident memory. Memory is measured per constructor subprocess and includes Python, NumPy, SciPy, and the optimizer; it is not isolated proof-array memory. The independent checker runs in a separate optimizer-free process. Table values are single-run descriptive timings, not statistical speed estimates. For the largest sparse program the exact checker is more expensive than the LP proposal. This cost is included rather than being hidden in publication overhead.

For the regret program, order the equality multipliers as s_{ti} for simplex equations and d_{ti} for regret equations. With $\ell_{ia} = \max_b r_{ib} - r_{ia}$, the reduced costs are

$$h_{tia} = \beta^t w_{ti} k_{ia} - s_{ti} + \ell_{ia} d_{ti}, \quad h_{ti}^D = -d_{ti} + \beta \sum_j P_{ji} d_{t-1,j},$$

where the preceding-date term is zero at $t = 0$. The lower certificate is

$$\max \left\{ 0, \sum_{ti} s_{ti} + \sum_{tia} \min(0, h_{tia}) + \varepsilon \sum_{ti} \min(0, h_{ti}^D) \right\}.$$

The upper certificate is the implementation Bellman evaluation of the serialized common policy. Regret is evaluated at every date and regime using exact rational arithmetic. Neither endpoint is inferred from an optimizer's success flag. Four mutation tests independently reject a negative probability, a forged upper endpoint, a forged lower endpoint, and an invalid transition row. Those tests are fault probes, not a completeness proof for the checker.

6. ALL TERMS OF THE CONTINUOUS TRANSFER BOUND

The density experiment uses $R = \max_{i,a} |r_{ia}|$, $K = \max_{i,a} k_{ia}$, and $G = 0$. Starting at the terminal date with zero error and zero payoff envelope, backward induction computes

$$a_t = \beta a_{t+1} + \beta \alpha_h B_{t+1}^J, \quad b_t = \beta b_{t+1} + \beta \alpha_h B_{t+1}^C, \quad B_t^J = R + \beta B_{t+1}^J, \quad B_t^C = K + \beta B_{t+1}^C.$$

The reported a_h and b_h are the maxima over dates. The two finite problems have allowances $\varepsilon + 2a_h$ and $\varepsilon - 2a_h$. The first supplies a lower endpoint; the second supplies a feasible lifted upper policy. Their independently checked arithmetic widths are η_+ and η_- . No finite solve is attempted when the tightening is nonpositive. The repair term is $D_C 4a_h / [(1 - \beta)(\varepsilon - 2a_h) +$

TABLE V
COMPUTED TERMS IN THE SUFFICIENT TRANSFER-WIDTH BOUND

ζ	η_+	η_-	$2b_h$	Repair bound	Total bound
0.001	2.232×10^{-8}	2.291×10^{-8}	0.058461	73.581005	73.639466
1e-06	2.291×10^{-8}	2.291×10^{-8}	0.000058	1.153667	1.153725

$4a_h]$. Every displayed total is computed from rational values, with decimal displays only for readability. The measured interval width in the main article is appreciably smaller than this sufficient worst-case bound. For $\zeta = 1/10$, a_h exceeds the permissible tightening threshold; that row is retained with its actual error budget and a failed sufficient condition. The exact structural reference is still available because its two aggregation identities hold at that parameter as well. This separates failure of a particular error bound from failure of the economic model or of the optimization problem.

7. AN EXACT TAIL BRIDGE FOR DISCOUNTED INFINITE-HORIZON REVISION

The main computational experiments are finite horizon. There is also a simple infinite-horizon extension with an explicit terminal trust boundary. Consider time-homogeneous bounded primitives, finite actions, and $0 < \beta < 1$. Let V^∞ be the exact operating fixed point and let a^* be a Borel stationary maximizing selector. Allow common Markov policies to be time dependent, and impose $J_t^p(x) \geq V^\infty(x) - \varepsilon$ for every date and state. Denote this infinite-horizon minimum by K_R^∞ . Let v_T be the finite-horizon revision minimum with terminal operating payoff V^∞ and zero terminal implementation cost.

PROPOSITION 1—Tail enclosure: *For any initial law and $\varepsilon \geq 0$,*

$$v_T \leq K_R^\infty \leq v_T + \frac{K\beta^T}{1 - \beta}.$$

Thus a finite certificate $[L_T, U_T]$ gives the infinite-horizon certificate $[L_T, U_T + K\beta^T/(1 - \beta)]$.

PROOF: Take an arbitrary infinite-horizon feasible policy. Replacing its continuation from date T by the operating-optimal selector weakly improves operating values at earlier dates. Its unchanged prefix is therefore feasible for the finite problem with terminal payoff V^∞ , and its truncated implementation cost does not exceed its full nonnegative cost. Taking infima proves the lower bound. Conversely, append a^* to any finite feasible prefix. This supplies an operating-feasible common Markov policy at all dates and adds at most $K\beta^T/(1-\beta)$ implementation cost. Approximate minimizers prove the upper bound without assuming a Borel function-space optimizer exists. *Q.E.D.*

This proposition does not give a stationary-policy optimum: the admissible prefix is time dependent. It also does not treat an approximate terminal value as exact, or an average-cost problem as discounted. A numerical use requires a certified terminal fixed point and tail policy, or additional witness errors. No infinite-horizon execution is claimed in the tables.

8. DATA-DEPENDENT REPAIR ACCOUNTING

For any two finite-horizon common policies p, q , define the implementation action values $Q_{k,t}^p(x, a) = k_t(x, a) + \beta P_t^a C_{t+1}^p(x)$. Their exact performance difference under the state law induced by q is

$$\int (C_0^q - C_0^p) d\nu = \sum_{t=0}^{T-1} \beta^t \mathbb{E}_q \left[\sum_a (q_t(a | X_t) - p_t(a | X_t)) Q_{k,t}^p(X_t, a) \right].$$

Subtract the two cost Bellman equations and iterate the resulting q -transition recursion to prove the identity. If $q_t = (1 - \alpha_t)p_t + \alpha_t \delta_{a_t}$, its additional cost is at most the discounted q -occupancy average of α_t times the action span of $Q_{k,t}^p$. For a given finite repaired policy this can be much sharper than $D_C \sup \alpha_t$. The generic stopping theorem still needs a uniform bound over unexplored policies; a favorable realized repair cost does not replace that requirement. The new structural cases use exact primal evaluation, and do not charge an unnecessary worst-case repair cost to their actual upper endpoints.

9. ALL SIGNED SAVINGS IN THE RETAINED PRIMARY COHORT

The next table is an exact re-tabulation of R40 rational endpoints. It does not change any policy or rerun the primary optimizer. A positive signed saving $L_D - U_R$ proves a gain over every feasible deterministic policy; a nonpositive value establishes no such separation. None of these signs establishes equality of L_R and U_R .

TABLE VI: All 42 uniform-initial primary certificates and signed deterministic savings

T	Installed rule	ε	L_R	U_R	$L_D - U_R$	Width (%)
4	defer	0.01	4.617419	4.730105	0.206584	2.38
4	defer	0.05	3.664042	3.890120	1.041948	5.81
4	neural31001	0.01	0.085102	0.089277	0.021898	4.68
4	neural31001	0.05	0.026223	0.030155	0.044177	13.04
4	neural31002	0.01	0.120480	0.125168	0.024289	3.75

T	Installed rule	ε	L_R	U_R	$L_D - U_R$	Width (%)
4	neural31002	0.05	0.047964	0.054026	0.058588	11.22
4	neural31003	0.01	0.094786	0.099085	0.022055	4.34
4	neural31003	0.05	0.031787	0.037227	0.047069	14.61
4	occupancy_stress	0.01	3.906911	4.020718	0.186083	2.83
4	occupancy_stress	0.05	3.020221	3.248042	0.941433	7.01
4	spline129	0.01	0.000000	0.000000	0.000000	0.00
4	spline129	0.05	0.000000	0.000000	0.000000	0.00
4	spline33	0.01	0.000000	0.000000	0.000000	0.00
4	spline33	0.05	0.000000	0.000000	0.000000	0.00
8	defer	0.01	10.503551	10.637940	0.172786	1.26
8	defer	0.05	9.449120	9.854394	0.896320	4.11
8	neural31001	0.01	0.343445	0.402312	0.066330	14.63
8	neural31001	0.05	0.104088	0.140458	-0.015946	25.89
8	neural31002	0.01	0.225489	0.277716	0.058701	18.81
8	neural31002	0.05	0.040296	0.059511	-0.015198	32.29
8	neural31003	0.01	0.552828	0.653902	0.211403	15.46
8	neural31003	0.05	0.086348	0.100439	0.244576	14.03
8	occupancy_stress	0.01	9.173627	9.365726	0.125235	2.05
8	occupancy_stress	0.05	7.983805	8.565982	0.682582	6.80
8	spline129	0.01	0.000000	0.000000	0.000000	0.00
8	spline129	0.05	0.000000	0.000000	0.000000	0.00
8	spline33	0.01	0.000000	0.000000	0.000000	0.00
8	spline33	0.05	0.000000	0.000000	0.000000	0.00
12	defer	0.01	15.317224	15.466905	0.145915	0.97
12	defer	0.05	14.142037	14.724122	0.747177	3.95
12	neural31001	0.01	2.770956	3.360150	0.133784	17.53
12	neural31001	0.05	1.584778	2.023532	0.379048	21.68
12	neural31002	0.01	2.051734	2.385046	0.151782	13.98
12	neural31002	0.05	1.195731	1.455145	0.309646	17.83
12	neural31003	0.01	0.772482	1.010276	0.217318	23.54
12	neural31003	0.05	0.093464	0.142107	-0.001624	34.23
12	occupancy_stress	0.01	13.420033	13.655059	0.044193	1.72
12	occupancy_stress	0.05	12.158413	12.877517	0.365025	5.58
12	spline129	0.01	0.000000	0.000000	0.000000	0.00
12	spline129	0.05	0.000000	0.000000	0.000000	0.00
12	spline33	0.01	0.000000	0.000000	0.000000	0.00
12	spline33	0.05	0.000000	0.000000	0.000000	0.00

10. THEOREM-TO-CHECKER MAP AND EVIDENCE CLASSIFICATION

Object	What is established	What is not inferred
Slack ladder	Analytic Borel proof; 48 exact tests with action-dependent three-state, three-action kernels, including $\beta = 1$	Finite tests are not a machine proof; the rate deteriorates with horizon.
Atomic aggregation	Two explicit kernel identities and a proof over arbitrary Borel policies; exact shear pushforward identity	Not a convergence theorem for arbitrary controlled atomic kernels or approximate lumpability.
Structured LP intervals	Independent rational probabilities, all-restart regrets, actual cost, and residual-corrected lower bounds for every new row	No reliance on solver success as a proof; no assertion about the continuum deterministic optimum.
Density transfer	Nonzero operator error and full propagated budget, plus independent finite certificates at both adjusted allowances	No assertion that every error budget is useful; the failed tightening row is retained.
R41 finest nonlinear audit	Complete independent scalar rebuild of the retained (32, 256, 1/2) object in the previously successful recorded run	Not a new upper policy, tighter operating tolerance, narrower optimum interval, or local re-run in R42.
Protocol and publication	Pre-execution case registration; content hashes and additive preservation checks	The source was not frozen in the case-registration commit; hashes do not verify a mathematical inequality.
Economic interpretation	Known-model adjustment costs and all-restart operating protection	No estimated primitives, empirical confidence sets, or monetary welfare estimates.

The inherited R41 audit reports 92,274,688 support-action inequalities and 2,097,152 state-date cells. It independently reproduces upper cost and lower bound

$$U = \frac{1667622841980339}{140737488355328}, \quad L = \frac{2497217525995195}{281474976710656},$$

with regret at most $2147483439/4294967296 < 1/2$. Its arithmetic replay took approximately 1,596 seconds. The manifest retains the source commit, verifier hash, proof hash, successful workflow identifier, and retrieved result. R42 does not misclassify this recorded prior run as a new local re-execution.

11. UNCHANGED EARLIER TECHNICAL MATERIAL

All earlier repository files remain intact. The separate file `HISTORY_R42.pdf` combines the complete R40 article and technical supplement without editing their pages; the latter already includes the R38 article and supplement. Revision-relative labels in that archive are historical. The current article and this supplement state R42's additions directly. The original primary and nonlinear experiments, failed sufficient certificates, adaptive local-LP work, model sensitivities,

and stopped-control derivations are not discarded. The stopped-control calculation is archival and is not included among the new completed targets.